A Stackelberg Game of Demand Response from the Aggregator's Perspective

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Introduction

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Demand Response

Demand Response (DR) is a policy that encourages consumers to shift their electricity consumption to times when the demand is lower.

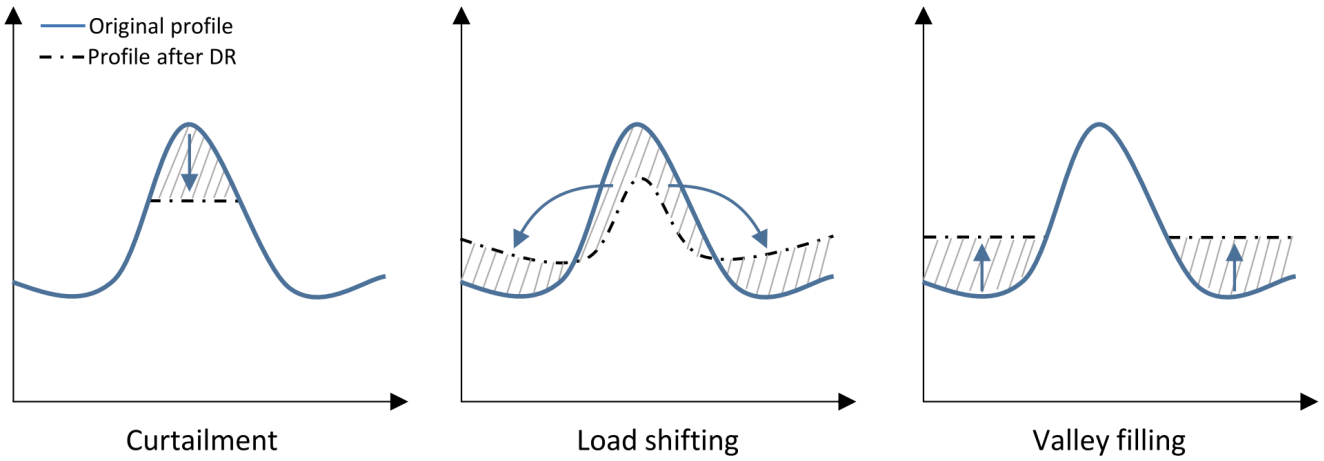

Figure 1: Types of DR [1].

Demand Response

DR is important because it helps

- energy efficiency
- stability of the grid
- reducing environmental impacts
- making profits

Stackelberg Game

Stackelberg Game

Stackelberg game is a strategic game between two groups of players. They are often called the leader and the followers.

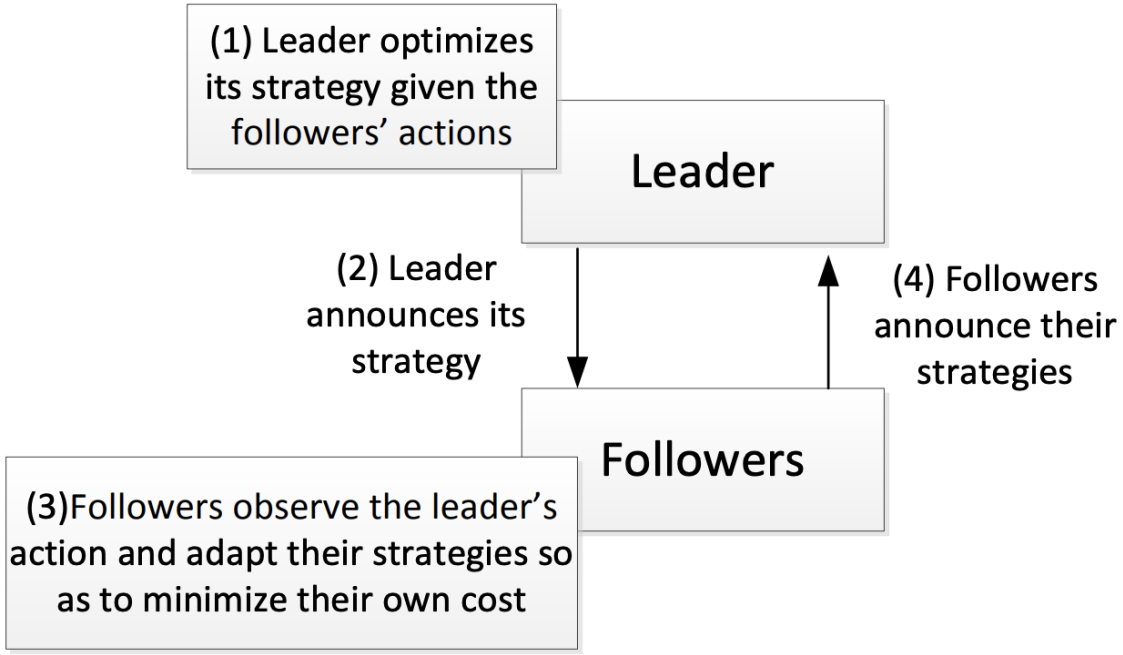

Figure 2: Stackelberg game structure.

Stackelberg Game

Mathematically,

$$
\begin{aligned}
\max \quad & \textit{objective function } 1 && \text{(1a)}\\
s.t. \quad & \textit{constraint } 1 && \text{(1b)}\\
& \textit{cosntraint } 2 && \text{(1c)}\\
& i \textit{ solves} \begin{cases} \max & \textit{objective function } 2 \\ s.t. & \textit{constraint } a \\ & \textit{constraint } b \end{cases} && \text{(1d)}
\end{aligned}
$$

Problem (1a) – (1c) are called leader's problem.

Problem (1d) is called the follower's problem.

The game is also known as Bilevel Optimization.

Stackelberg Game

First introduced by Heinrich F. von Stackelberg in 1934, this game considers the hierarchical structure of decision-making processes in real-world scenarios.

- Strategic Analysis: It provides insights into competitive behavior in markets where firms can influence each other's decisions.
- Policy Implications: Helps policymakers understand the impact of regulation on market dynamics and competition.
- Optimization: Used in various fields such as economics, business strategy, environmental policy, and supply chain management to optimize decisions based on hierarchical structures.

Literature

The Stackelberg game has been recently applied in DR studies to model the interactions between stakeholders in energy systems.

- Sanguansuttigul et al. [2] proposed a DR model using bilevel optimization to maximize the profit for an electricity provider and minimize the electricity bill for consumers.
- Tiwari et al. [3] proposed a DR model using Stackelberg game to minimize the operation cost for a distribution system operator (DSO) while maximizing incentives of the load aggregator.

Rationales & Objectives

From the literatures...

- They mainly focused on the benefits of the load aggregator and consumers separately.
- Those models are price-based DR programs.

Novelty

We consider both load aggregator and consumers in one model from the load aggregator's perspective in a DR program with a special mission of load reduction during on-peak period.

Objectives

- DR model using Stackelberg game to analyze the interactions between the load aggregator and the consumers.
- How can this proposed model help the load aggregator to choose the participants, save bills and obtain commission?

Framework and Methodology

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Framework

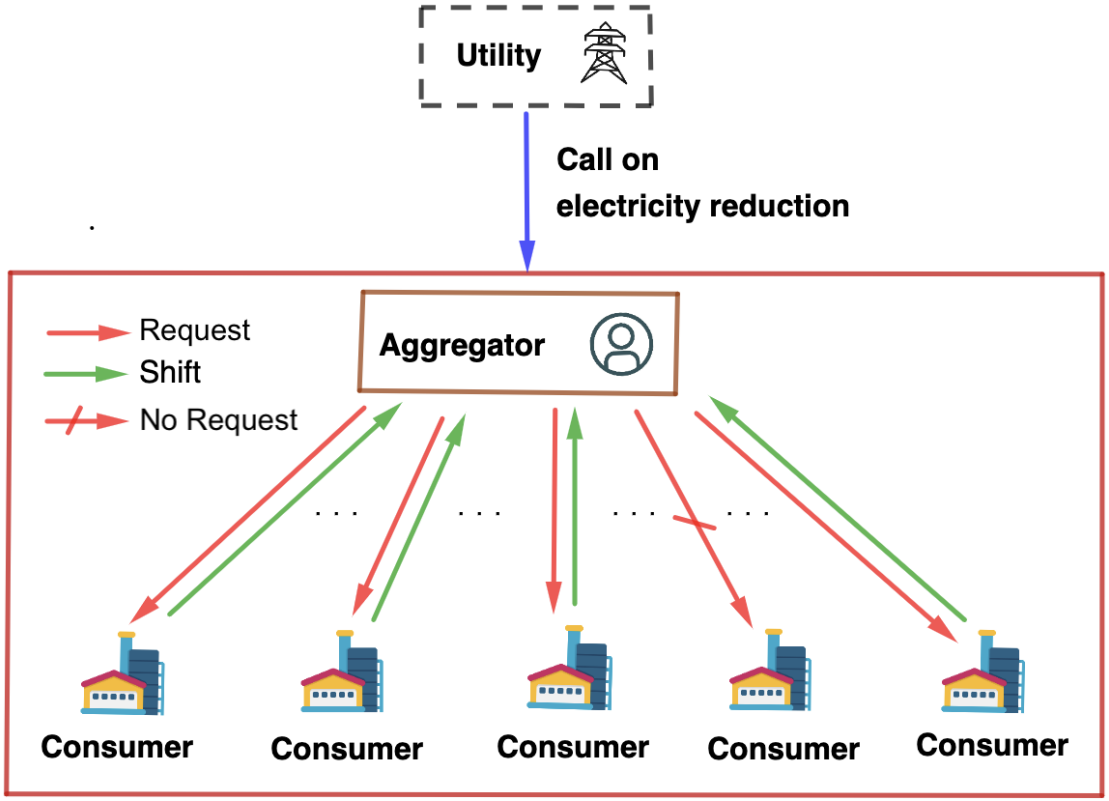

Figure 3: Framework of the study.

Framework

- Players in the framework: Aggregator and consumers.
- Consumers can be residential, commercial, or industrial.
- Aggregator is given a mission to reduce the high consumption during on-peak hours.
- It then calls on the consumers to participate in the DR program by sending requests of load reduction.
- Each consumer then decides how much load to reduce based on the request, their baseline consumption, and the incentives provided by the aggregator.

Stackelberg game

The complete formulation is

$$\begin{cases} \max\limits_{x_i} & \textit{Total load reduction} + \textit{fairness} + \textit{commission} & (2) \\ \text{s.t.} & \textit{Total load reduction} = \textit{the target amount} & (3) \\ & \textit{Request} \leq \textit{baseline} & (4) \\ & y_i \in \mathcal{S}(x_i) & (5) \end{cases}$$

where $\mathcal{S}(x_i)$ is the solution set of the follower i's problem (6)–(8).

$$\begin{cases} \min\limits_{y_i} & \textit{Payment} + \textit{dissatifaction} - \textit{reward} & (6) \\ \text{s.t.} & \textit{shift} \leq \textit{request} & (7) \\ & \textit{percentage of reduction} \in [0, 1]. & (8) \end{cases}$$

Note:

x_i is a decision variable of electricity request from the leader side.

y_i is a decision variable of shifted load of consumer i in percentage.

Stackelberg game

Solva bility

In solving the proposed model, we transform it into a single-level optimization programming using KKT conditions [4, 5]. Then we solve it using `Julia programming language with JuMP package and Gurobi solver.`

Results and Discussions

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Setups

In the simulations,

- We consider 10 consumers with different load baseline consumptions based on the literatures.
- The reduction targets are set to 800 kWh and 1 500 kWh.
- There are different fairness weights to see their impacts on the decision-making of the load aggregator.
- The time of a day is divided into on-peak and off-peak hours as

Time	00h-09h00	09h00-21h00	21h00-24h00
types	off-peak	on-peak	off-peak

Table 1: On-peak and off-peak hours per a day

Load aggregator's side

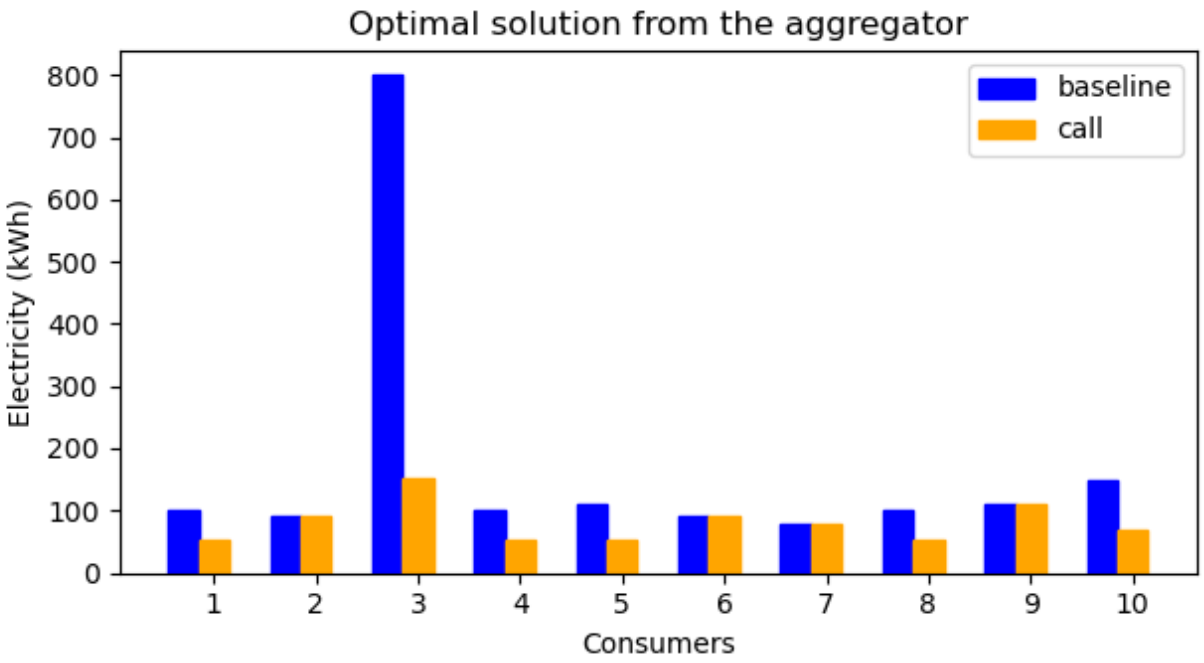

Figure 4: Optimal decision from the aggregator

- Consumer 2,6,7 and 9 are called to shift the whole operation to off-peak hours.
- Total reduction is 84%.
- The commission is proportional to the amount of shifted load.

Consumers' side

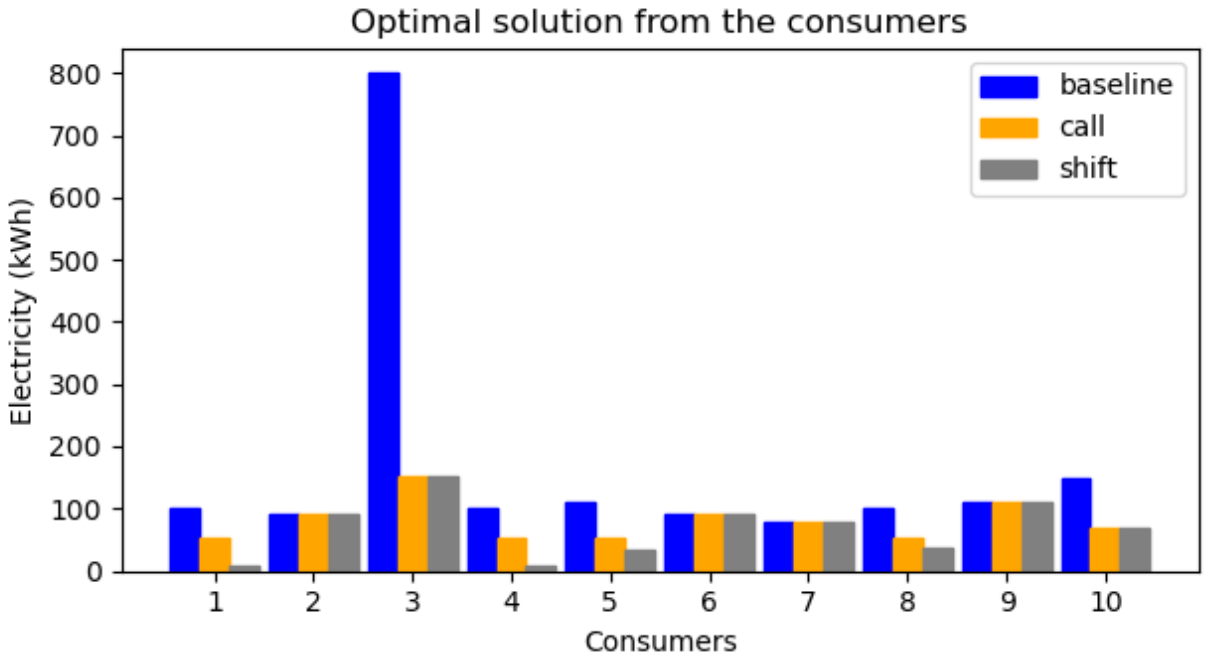

Figure 5: Optimal decision from the consumers

- Consumers 2,3,6,9 and 10 fully comply with the request, then they recieve highest rewards.
- Consumer 2 saves the bill by 44%, consumer 3 saves 31%.
- Consumer 1 is called to reduce 52.4 kWh, yet he manages to reduce 7.55 kWh (14.4%), less reduction as consumer 4 $\Rightarrow$ less reward.

Conclusion

- We proposed an innovative Stackelberg game model to study the interactions between a load aggregator and end-consumers in a DR program with a mission of load reduction during on-peak hours.
- The results show that the proposed model can effectively achieve the load reduction target and benefit to both stakeholders while considering fairness among consumers and providing incentives for their participation.
- The proposed model assists the load aggregator choose the consumers to participate in the program.

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Thank you for your attention!